How Many Hidden Units Is a Skip Connection Worth?

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Abstract

Residual networks add the input of every block to its output, and skipless networks can now be trained to match them. Can a skip let a block compute a function that no skipless block of the same kind computes? Here we show that, in blocks without biases, a skip is worth exactly d_{model} hidden ReLU or GELU units for almost every block, with d_{model} the model width, and more than any number of ReLU^2, gated or $\tanh$ units or attention heads. Between a linear embedding and unembedding, residual networks of ReLU^2, gated or $\tanh$ blocks equal no skipless network, and a skipless ReLU network needs $2d_{\text{model}}$ units in its first block to equal almost any residual one. Without normalization, every skipless network is still a limit of residual networks of the same widths. These results say when a residual block can be rewritten without its skip.

1 Introduction

Almost every deep network trained today adds the input of a block to its output. A residual block computes $x + B(x)$ (He et al., 2016), and transformers stack such blocks (Vaswani et al., 2017). The usual reason is training: the sum gives the gradient a short path through the network, and it makes the identity map easy to express (Hardt and Ma, 2017; Veit et al., 2016).

Skipless networks now train too. He and Hofmann (2024) remove the skips of transformer blocks without losing training speed, and Ji et al. (2026) train a skipless ViT-B on ImageNet that outperforms the residual one. Can a skip let a block compute a function that no skipless block of the same kind computes?

We answer this question block by block. Take an MLP block B with m hidden units, or an attention block with H heads, on hidden states of width d_{model}. Is the residual block $x \mapsto x + B(x)$ also a block of the same kind without a skip, of the same width or wider? If it is, the skip is worth some hidden units; if not, it is worth more than any number of them.

Without biases, the answer depends on the activation.

- For ReLU and GELU blocks, a skip is worth exactly d_{model} units. Since $\sigma(z) - \sigma(-z) = z$, d_{model} extra units absorb it, and for almost every block fewer do not (Theorem 3.1(ii)).
- For ReLU^2, gated and $\tanh$ blocks, no number of units absorbs a skip (Theorem 3.2).
- For attention blocks, no number of heads absorbs a skip (Theorem 3.3(i)).

These answers hold for blocks without biases, as in many current language models (Team OLMo et al., 2025; Yang et al., 2025). With biases, ReLU^2 and gated blocks absorb a skip too (Section 2).

Four one-line reasons. Follow the input along a ray, $x = tu$ with $t > 0$. The skip grows like t. A ReLU^2 block grows like t^2, a gated block starts at order t^2 near the origin, and a $\tanh$ block stays bounded, so none of them can produce the linear term (Figure 1). Attention mixes the tokens of a sequence, while a skip leaves each token where it is.

*Author contributions. Marko Karbevski (MK) conceived and led the study. Antonij Mijoski (AM) proved the absorption result for a single ReLU block (Theorem 3.1(i), Theorem A.5), the difficult case, paving the way forward. MK extended the tools to GELU (Theorem A.7), proved the cases of ReLU^2 and of the gated (GLU) variants (Theorems 3.2 and 4.1) and the results at depth (Theorem 4.2 and Proposition 4.3). Both authors worked on the writing. All of MK's work on the results and the proofs was done with AI assistants.

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Proven	**ReLU**	**GELU**	**tanh**	**ReLU^2, ReGLU**	**SwiGLU, GeGLU**	**Attention**
one block, same width	measure zero Thm. 3.1(i)		never Thm. 3.2			never Thm. 3.3(i)
one block, more units	exactly d_{model} more Thm. 3.1(ii)		no number Thm. 3.2			no number of heads Thm. 3.3(i)
networks of one block, same width	measure zero Thm. 3.1(i)		never Thm. 4.1(ii)	never Thm. 4.1(i)		never Thm. 3.3(ii)
deeper networks, same width	measure zero if the first block has fewer than $2d_{\text{model}}$ units Thm. 4.2	–	never Thm. 4.1(ii)	never Thm. 4.1(i)		–
networks, more units per block	d_{model} more suffice Prop. 4.3		no number Thm. 4.1(ii)	no number Thm. 4.1(i)		–
networks with pre-RMSNorm, $\varepsilon > 0$	–	–	–	never Thm. 4.1(i)		–
networks with pre-RMSNorm, $\varepsilon = 0$	measure zero, at any width Prop. 4.4(iii)					
skipless as limits of residual	yes, at the same widths Prop. 5.1(i)					
with pre-RMSNorm, $\varepsilon = 0$	yes, if no hidden state has a zero row Prop. 5.1(ii)					
with pre-RMSNorm, $\varepsilon > 0$	–					

Table 1: Can a skipless network equal a residual one? Never: for no weights and no width. Measure zero: only for weights in a set of measure zero. Exactly d_{model} more: d_{model} more units suffice, and for almost every block fewer do not. A dash: open here. No biases; notation in Section 2.

Networks. Throughout, a network starts with a linear embedding and ends with a linear unembedding, which reads the last hidden state through an RMSNorm in normalized networks. The picture survives in networks, with one twist for ReLU: a skipless ReLU network needs $2d_{\text{model}}$ units in its first block to equal almost any residual one (Theorem 4.2). Under RMSNorm with $\varepsilon = 0$, a skipless network forgets the length of its input and the skip keeps it, so no width helps (Proposition 4.4(iii)).

Exact, yet approximable. Without normalization, every skipless network is a limit of residual networks of the same widths (Proposition 5.1(i)): the skip adds functions and, in the limit, removes none. Table 1 collects the results; Section 2 fixes the setting, Section 3 treats one block, Section 4 networks and Section 5 limits. Proofs are in the appendix.

2 Setup

One equation states the question: can a block absorb a linear term?

Blocks. A hidden state is a matrix $h \in \mathbb{R}^{T \times d_{\text{model}}}$ whose rows are the T tokens of a sequence. An MLP block acts on each token separately. On a row $x \in \mathbb{R}^{1 \times d_{\text{model}}}$ it computes

$$B_\theta(x) = f(xW_{\text{read}})\, W_{\text{write}}, \qquad \theta = (W_{\text{read}}, W_{\text{write}}), \tag{1}$$

with a read W_{read} and a write W_{write}. In a vanilla block with m hidden units, $W_{\text{read}} \in \mathbb{R}^{d_{\text{model}} \times m}$, $W_{\text{write}} \in \mathbb{R}^{m \times d_{\text{model}}}$, and f applies an activation σ to each unit: ReLU, GELU, $\tanh$ or ReLU^2, $\sigma(z) = \max(z, 0)^2$. In a gated block the read has two halves, $W_{\text{read}} = [\, W_{\text{g}} \mid W_{\text{v}} \,]$, and unit i computes $\sigma\big((xW_{\text{g}})_i\big)\,(xW_{\text{v}})_i$, with σ the ReLU in ReGLU, SiLU in SwiGLU and GELU in GeGLU (Shazeer, 2020). Attention blocks follow in Section 3.3.

The question as an equation. A skip adds x to the output of a block. More generally, can a block absorb a linear term xM, for a $d_{\text{model}} \times d_{\text{model}}$ matrix $M \neq 0$? We look for parameters θ' of a block

with the same activation, and of any width, such that

$$xM + B_\theta(x) = B_{\theta'}(x) \qquad \text{for all } x \in \mathbb{R}^{1\times d_{\text{model}}}. \tag{2}$$

The skip is $M = I$. Other matrices cover learned linear skips and, below, networks.

Networks. Every network starts with a linear embedding E and ends with a linear readout D, the unembedding: it maps an input x to $h_0 = xE$, passes it through L blocks, and returns $y = h_L D$. A *skipless* network has $h_\ell = B_\ell(h_{\ell-1})$, and a *residual* network has $h_\ell = h_{\ell-1} + B_\ell(h_{\ell-1})$. In a *pre-RMSNorm* network every block and the readout read the normalized state $N(h)$, so that $y = N(h_L)D$. Here N divides each row of h by $\sqrt{\|h\|^2/d_{\text{model}} + \varepsilon}$, where $\|h\|$ is the norm of that row and $\varepsilon \geq 0$.

Scope. Blocks have no biases, and the answers depend on it. With biases, $4d_{\text{model}}$ ReLU^2 units absorb a skip, since $\frac{1}{4}\big(r(z+1) - r(z-1) - r(1-z) + r(-1-z)\big) = z$ for $r(z) = \max(z, 0)^2$. So do $2d_{\text{model}}$ gated units whose values are constant biases, since the gates satisfy $\sigma(z) - \sigma(-z) = z$. For simplicity, every block also has at least d_{model} hidden units, all MLP blocks of a network are of one kind, all attention blocks of a network have the same number of heads, and inputs and outputs have dimension at least $d_{\text{model}} \geq 2$. Most results hold more generally, and the appendix states them so. *Generic* weights, or almost every choice of weights, are those outside a set of Lebesgue measure zero.

3 One block

For a ReLU or GELU block a skip is worth exactly d_{model} hidden units. For ReLU^2, gated, $\tanh$ *and attention blocks it is worth more than any number.*

3.1 ReLU and GELU blocks: a skip is worth d_{model} units

Both activations satisfy

$$\sigma(z) - \sigma(-z) = z, \tag{3}$$

so a unit and its mirror image, with the opposite read and the opposite write, add up to a linear term. This gives two ways to absorb xM.

With more units. Let unit i have read w_i and write a_i. A new unit with read $-w_i$ and write $-b$, together with the write $a_i + b$ for unit i, adds $(xw_i)\, b$ to the block, by (3). So each new unit adds one rank-one term, and d_{model} new units, mirroring d_{model} units whose reads span $\mathbb{R}^{d_{\text{model}}}$, add any xM.

At the same width. Flipping the read of unit i to $-w_i$ changes the block by $-(xw_i)\, a_i$. So a block of the same width absorbs xM if some set S of units has $W_{\text{read}}[:, S]\, W_{\text{write}}[S, :] = -M$, where $W_{\text{read}}[:, S]$ and $W_{\text{write}}[S, :]$ keep the columns and rows of the units in S.

The theorem says that these are the only ways. A block is *nondegenerate* if the columns of W_{read} are nonzero and pairwise non-collinear and the rows of W_{write} are nonzero.

Theorem 3.1 (ReLU and GELU blocks)**.** *Let σ be ReLU or GELU.*
(i) *If B_θ is nondegenerate, a block $B_{\theta'}$ of the same width satisfies (2) if and only if some set S of hidden units has $W_{\text{read}}[:, S]\, W_{\text{write}}[S, :] = -M$; for generic θ there is none.*
(ii) *If M is invertible and W_{read} has rank d_{model}, a block $B_{\theta'}$ with d_{model} more hidden units satisfies (2), and for generic θ none with fewer does.*

Proof in Appendix A.1.

In (i), generic weights give no set of units with the required product, since there are finitely many sets and each gives one matrix. By (ii), the skip, $M = I$, is worth exactly d_{model} units.

The proof matches the units of the two blocks. The even part of the activation, $\big(\sigma(z) + \sigma(-z)\big)/2$, fixes the units of the absorbing block up to sign and scale: through the kinks of $|z|$ for ReLU, and through the linear independence of ridge functions for GELU. The odd part, $z/2$, turns each sign flip into a linear term, as Petzka et al. (2020, Section 3) show for two-layer ReLU networks with one output. Each extra unit then adds at most one rank-one term, and $M = I$ has rank d_{model}.

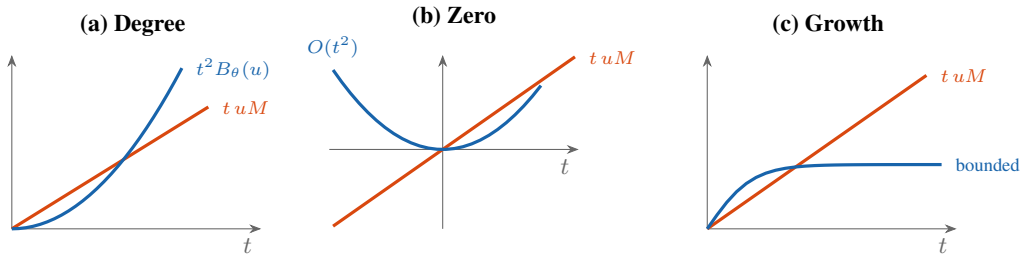


Figure 1: Why no width absorbs the skip. Along a ray $x = tu$ the skip grows like t. (a) A homogeneous block of degree 2 grows like t^2. (b) A gated block is of order t^2 near $t = 0$. (c) A tanh block stays bounded.

Networks of one block. A network of one block computes $x \mapsto xED + f(xEW_{\text{read}})\,W_{\text{write}}D$. Theorem 3.1(i) applies with the reads EW_{read}, the writes $W_{\text{write}}D$ and $M = ED$, so a skipless network of the same width equals it only for weights in a set of measure zero.

3.2 ReLU^2, gated and tanh blocks

Recall that a block absorbs xM if a block of the same kind satisfies (2): $xM + B_\theta(x) = B_{\theta'}(x)$ for all x.

Theorem 3.2 (No width suffices)**.** *Let B_θ be a* tanh *block, a block that is positively homogeneous of degree $k \neq 1$, such as a ReLU^2 or ReGLU block, or a gated block whose gate vanishes at 0 and is differentiable there, such as a SwiGLU or GeGLU block. No block $B_{\theta'}$ of the same kind and of any width satisfies (2).*

Proof in Appendix A.1.

The three cases of Figure 1 are three one-line proofs. A homogeneous block satisfies $B_\theta(tu) = t^k B_\theta(u)$ for $t > 0$, so along a ray (2) reads $t\,uM = t^k\big(B_{\theta'}(u) - B_\theta(u)\big)$, which forces $uM = 0$ for every u when $k \neq 1$. A gate that vanishes at 0 makes every unit of order t^2 near the origin, so both blocks have derivative 0 there, while xM has derivative M. A tanh block is bounded, and xM is not. A linear block, by contrast, absorbs any skip: $xM + xA = x(M + A)$.

3.3 Attention blocks

Attention mixes tokens, and a skip keeps each token where it is. Two inputs make this precise. If all tokens are equal, mixing changes nothing, so the value–output products of the two blocks must differ by M. If one token is zero, its query is zero, and attention gives it an equal mix of all tokens: the difference of the two blocks gives it the mean token times M, while the skip gives it zero. So $M = 0$.

An *attention block* with parameters $\theta = (W_{Q,i}, W_{K,i}, W_{V,i}, W_{O,i})_{i=1}^{H}$ maps $h \in \mathbb{R}^{T \times d_{\text{model}}}$, whose rows are the $T \geq 2$ tokens, to

$$B_\theta(h) = \sum_{i=1}^{H} P_{\Gamma_i}(h)\, h\, W_{V,i} W_{O,i}, \qquad P_\Gamma(h) = \operatorname{softmax}\big(h\,\Gamma\,h^\top\big), \qquad \Gamma_i = W_{Q,i} W_{K,i}^\top / \sqrt{d_h}, \tag{4}$$

where $W_{Q,i}, W_{K,i}, W_{V,i} \in \mathbb{R}^{d_{\text{model}} \times d_h}$ and $W_{O,i} \in \mathbb{R}^{d_h \times d_{\text{model}}}$. The softmax acts on each row of the $T \times T$ matrix of scores, with or without a causal mask, so each row of $P_\Gamma(h)$ is a probability vector. The read of the block holds the matrices $W_{Q,i}$, $W_{K,i}$ and $W_{V,i}$, and its write the matrices $W_{O,i}$. Head i has the score matrix Γ_i and the value–output product $W_{V,i}W_{O,i}$.

Theorem 3.3 (Attention blocks)**.** *Let θ and θ' be parameters of attention blocks with any numbers and widths of heads, and let $M \neq 0$ be a $d_{\text{model}} \times d_{\text{model}}$ matrix.*
(i) *For almost every $h \in \mathbb{R}^{T \times d_{\text{model}}}$, $hM + B_\theta(h) \neq B_{\theta'}(h)$.*
(ii) *With an embedding E and a readout D with $ED \neq 0$, the residual network $x \mapsto \big(xE + B_\theta(xE)\big)D$ differs from every skipless network $x \mapsto B_{\theta'}(xE')D'$.*

A special case of Theorem A.12, proved in Appendix A.2.

The proof gives a gap: on some input with rows of norm at most 1, one row of $hM + B_\theta(h) - B_{\theta'}(h)$ has norm at least $\|M\|/3$, with $\|M\|$ the spectral norm (Theorem A.12(ii)). It uses only that each row of attention weights sums to 1 and is uniform for a zero query, so it also holds with rotary position embeddings. With three or more tokens, the heads of the two blocks must even cancel in groups with the same score matrix (Proposition A.14 in Appendix A.2).

4 Networks

Between an embedding and a readout the picture survives: residual networks of ReLU^2, gated or $\tanh$ *blocks equal no skipless network, and a skipless ReLU network needs a wide first block.*

Recall that a network embeds its input as $h_0 = xE$ and reads out $y = h_L D$, and that its skips add $h_{\ell-1}$ at every block.

Theorem 4.1 (Networks that never absorb their skips)**.** *Let the residual network have $ED \neq 0$.*
(i) *For gated blocks whose gate vanishes at 0 and is differentiable there, and for blocks that are positively homogeneous of degree $k > 1$, no skipless network of any depth and widths equals the residual network, also under pre-RMSNorm with $\varepsilon > 0$.*
(ii) *Without normalization, the same holds for* $\tanh$ *blocks.*

Proof in Appendix A.1.

With $E = D = I$, this covers compositions of blocks. The reasons are those of Figure 1, now for the whole network. A gated block, and a homogeneous block of degree $k > 1$, has derivative 0 at the origin, so every skipless network of them has Jacobian 0 at $x = 0$, while a residual network has Jacobian $ED \neq 0$ there; this comparison of derivatives is why the networks need $k > 1$. A skipless $\tanh$ network is bounded, while a residual one grows like xED along rays.

ReLU networks need a wide first block.

Theorem 4.2 (ReLU networks)**.** *Consider networks of ReLU blocks. For almost every choice of weights, a residual network equals no skipless network whose first block has fewer than $2d_{\text{model}}$ hidden units, whatever its depth and its other widths.*

Proof in Appendix A.1.

The proof compares ranks. Near almost every input, a generic residual ReLU network is a linear map of rank d_{model}. No unit of the first block is active at both x and $-x$, so with fewer than $2d_{\text{model}}$ units a skipless network has fewer than d_{model} active units at one of the two inputs, and some direction of inputs leaves it unchanged there.

Conversely, the construction of Section 3.1 works block by block.

Proposition 4.3 (Extra units)**.** *Let σ be ReLU or GELU. A residual network whose reads all have rank d_{model} equals a skipless network with d_{model} more hidden units in every block.*

Proof in Appendix A.1.

So Ld_{model} extra units in all suffice; for ReLU, the first block needs at least $2d_{\text{model}}$ units (Theorem 4.2).

RMSNorm with $\varepsilon = 0$: the skip keeps the length of the input. With $\varepsilon = 0$, $N(ch) = N(h)$ for every $c > 0$, so the first normalization of a skipless network discards the length of each input token. The skip of a residual network passes it on.

Proposition 4.4 (Absorption under RMSNorm with $\varepsilon = 0$)**.** *Consider pre-RMSNorm networks with $\varepsilon = 0$, with blocks of any kind and width.*
(i) *The function F of every skipless network satisfies $F(cx) = F(x)$ for all $c > 0$.*
(ii) *A residual network with this property computes $x \mapsto N(xE)D$.*
(iii) *The residual networks that equal a skipless network have parameters in a set of measure zero.*

Proof in Appendix A.3.

For (ii), let $c \to \infty$: the blocks read $N(h)$, whose rows have length $\sqrt{d_{\text{model}}}$, so their outputs stay bounded while the skips carry cxE, and the output tends to $N(xE)D$.

5 Skipless networks are limits of residual ones

A skip removes no function in the limit: without normalization, residual networks of the same widths approach every skipless network.

Proposition 5.1 (Skipless networks as limits of residual networks)**.** *Let F be a skipless network whose blocks are locally Lipschitz.*
(i) Without normalization. *The network F is a limit of residual networks with blocks of the same kinds and widths, uniformly on bounded sets of inputs.*
(ii) Under pre-RMSNorm with $\varepsilon = 0$. *If no hidden state of F has a zero row at any input x whose rows of xE are nonzero, the same holds on bounded sets of such inputs.*

Proof in Appendix A.4.

Blocks that outweigh their skips. The proof scales the reads of block ℓ by $t^{1-\ell}$, its writes by t^{ℓ} and the readout by t^{-L}. Block ℓ then sees the hidden state of F, up to $O(1/t)$, and outweighs its skip by a factor t, so the residual network computes $F + O(1/t)$ (Remark A.15). The large constants sit in the writes, so no activation sees a large argument, and the construction applies alike to $\tanh$, GELU, ReLU, ReLU^2, gated and attention blocks. The weights diverge as $t \to \infty$: the proposition says which functions residual networks reach in the limit, and leaves open how to parametrize them for training.

Under RMSNorm. With $\varepsilon = 0$, RMSNorm ignores the size of the hidden states, so only the writes grow. With $\varepsilon > 0$ the construction fails, since on a hidden state of size t^{ℓ} RMSNorm acts as RMSNorm with $\varepsilon/t^{2\ell}$; whether skipless networks are limits of residual ones there is open.

6 Related work

Expressivity of residual networks. Lin and Jegelka (2018) show that residual ReLU networks with one hidden unit per block are universal approximators, which networks of width d_{in} without skips are not; they count the identity map as d_{model} hidden units, the view we take throughout. Hardt and Ma (2017) show that deep linear residual networks have no spurious local optima, and that residual ReLU networks can fit any finite sample once they have more parameters than samples. Both results concern approximation or finite samples; ours concern exact equality of functions, block by block and activation by activation.

Identifiability. For $\tanh$ networks, the function determines the weights up to sign flips and permutations of hidden units (Fefferman, 1994; Sussmann, 1992), and similar statements hold for other activations (Vlačić and Bölcskei, 2022) and for ReLU networks (Petzka et al., 2020; Phuong and Lampert, 2020; Rolnick and Kording, 2020). Our proofs identify the units of a block from its function in the same spirit, and then ask what is left for the linear term of the skip.

Training without skips. Skipless networks train with careful initialization or activation shaping (Martens et al., 2021; Zhang et al., 2022), and skipless transformers with modified attention (He et al., 2023; Noci et al., 2023) or simplified blocks (He and Hofmann, 2024). Ji et al. (2026) train a skipless ViT-B that outperforms the residual one. Our results say how many units such a network needs to compute exactly what a residual network computes.

7 Conclusion

Can a skip let a block compute a function that no skipless block of the same kind computes? Without biases, yes: for almost every block, no skipless block of the same kind and width computes $x + B(x)$. ReLU and GELU blocks need exactly d_{model} more units, and ReLU^2, gated, $\tanh$ and attention blocks cannot absorb the skip at any width. Between an embedding and a readout the same holds for ReLU^2, gated and $\tanh$ networks, a skipless ReLU network needs a first block of $2d_{\text{model}}$ units, and under RMSNorm with $\varepsilon = 0$ no width suffices. So, without normalization, a trained residual ReLU

or GELU network can be rewritten exactly without skips with d_{model} more units per block, and a ReLU^2, gated or $\tanh$ network cannot be rewritten at any width.

Open questions. The dashes of Table 1 leave four questions.

(1) GELU and attention networks at depth, and GELU, $\tanh$ and attention networks under RMSNorm with $\varepsilon > 0$. At the same widths, a count of functional dimensions answers them in a companion paper (Karbevski and Mijoski, 2026); with more units they are open.
(2) ReLU networks whose first block has at least $2d_{\text{model}}$ units, and ReLU networks under RMSNorm with $\varepsilon > 0$.
(3) Skipless networks as limits of residual networks under RMSNorm with $\varepsilon > 0$.
(4) Approximate absorption: how many units a skip is worth at a given error.

Use of AI

The conception of the study is the authors' own. One author obtained their results, proofs and experiments working with AI assistants, through continuous conversation; the assistants were used to find, verify and simplify proofs and to write the code of the experiments, which the authors validated.

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A Proofs

A.1 MLP blocks and networks (Sections 3 and 4)

We recall Theorems 3.1 and 3.2 from Section 3 and Theorems 4.1 and 4.2 and Proposition 4.3 from Section 4, for the question (2). Below we write B for B_θ, and B' for $B_{\theta'}$ with $\theta' = (W'_{\mathrm{read}}, W'_{\mathrm{write}})$.

Theorem 3.1 (ReLU and GELU blocks)**.** *Let σ be ReLU or GELU.*
(i) *If B_θ is nondegenerate, a block $B_{\theta'}$ of the same width satisfies (2) if and only if some set S of hidden units has $W_{\mathrm{read}}[:, S]\, W_{\mathrm{write}}[S, :] = -M$; for generic θ there is none.*
(ii) *If M is invertible and W_{read} has rank d_{model}, a block $B_{\theta'}$ with d_{model} more hidden units satisfies (2), and for generic θ none with fewer does.*

Theorem 3.2 (No width suffices)**.** *Let B_θ be a $\tanh$ block, a block that is positively homogeneous of degree $k \neq 1$, such as a ReLU^2 or ReGLU block, or a gated block whose gate vanishes at 0 and is differentiable there, such as a SwiGLU or GeGLU block. No block $B_{\theta'}$ of the same kind and of any width satisfies (2).*

Theorem 4.1 (Networks that never absorb their skips)**.** *Let the residual network have $ED \neq 0$.*
(i) *For gated blocks whose gate vanishes at 0 and is differentiable there, and for blocks that are positively homogeneous of degree $k > 1$, no skipless network of any depth and widths equals the residual network, also under pre-RMSNorm with $\varepsilon > 0$.*
(ii) *Without normalization, the same holds for $\tanh$ blocks.*

Theorem 4.2 (ReLU networks)**.** *Consider networks of ReLU blocks. For almost every choice of weights, a residual network equals no skipless network whose first block has fewer than $2d_{\mathrm{model}}$ hidden units, whatever its depth and its other widths.*

Proposition 4.3 (Extra units)**.** *Let σ be ReLU or GELU. A residual network whose reads all have rank d_{model} equals a skipless network with d_{model} more hidden units in every block.*

Overview. Theorem 3.2 compares orders. Along rays from the origin a linear skip scales like λ and a block that is homogeneous of degree $k \neq 1$ like λ^k, and near the origin a gated block is of second order, so the two sides of the absorption equation cannot agree (Propositions A.2 and A.3); comparing Jacobians at the origin extends this to networks of any depth, Theorem 4.1 (Corollary A.4). Theorem 3.1(i) is the substantial case. The absorption equation splits into its odd and even parts in

x. The even part determines the hidden units of the absorbing block up to sign, and for ReLU up to a positive rescaling, through the kinks for ReLU (Theorem A.5) and through linear independence of ridge functions for GELU (Lemma A.6 and Theorem A.7). The odd part of ReLU and GELU is exactly $z/2$, so the odd part of the equation is linear and reads off the set S of units whose signs flip; conversely, flipping the signs of the units in S absorbs the skip whenever the condition holds. The generic clause holds because the condition is a finite union of algebraic sets of positive codimension (Corollary A.10), and Remark A.9 covers singular M. Theorem 3.1(ii) counts the extra units that the matching requires (Proposition A.11). Proposition 4.3 applies Theorem 3.1(ii) block by block. Theorem 4.2 compares the ranks of the linear pieces at an input and at its negative.

The arguments show that the function of a block determines its parameters, in the manner used for $\tanh$ networks by Sussmann (1992) and Fefferman (1994) and for ReLU networks by Phuong and Lampert (2020), Petzka et al. (2020) and Rolnick and Kording (2020): kink alignment for ReLU, ridge-function independence for GELU. For two-layer ReLU networks with a single output, Petzka et al. (2020, Section 3) already give the corresponding sign flips, show that irreducible networks computing the same function differ only by these flips, permutations and positive rescalings (their Theorem 1), and state the matching criterion for removing a linear term. Lemma A.1 reduces any invertible skip to the identity skip.

Setting. Throughout, $W_{\text{read}} \in \mathbb{R}^{d_{\text{model}} \times m}$ and $W_{\text{write}} \in \mathbb{R}^{m \times d_{\text{model}}}$, σ acts coordinatewise, w_i is column i of W_{read} and $\hat{w}_i$ is row i of W_{write}, and inputs x are row vectors. For $S \subseteq \{1, \dots, m\}$, $W_{\text{read}}[:, S]$ is the submatrix of columns in S, $W_{\text{write}}[S, :]$ the submatrix of rows in S, and Π_S the diagonal projector with $(\Pi_S)_{ii} = 1$ exactly when $i \in S$, so that $W_{\text{read}} \Pi_S W_{\text{write}} = W_{\text{read}}[:, S] W_{\text{write}}[S, :]$. A family of vectors is *pairwise non-collinear* if no one of them is a scalar multiple of another, and a block is *nondegenerate* if the columns of W_{read} are nonzero and pairwise non-collinear and the rows of W_{write} are nonzero. The question is whether there is a block $B'(x) = \sigma(xW'_{\text{read}})W'_{\text{write}}$ of the same width, with $W'_{\text{read}} \in \mathbb{R}^{d_{\text{model}} \times m}$ and $W'_{\text{write}} \in \mathbb{R}^{m \times d_{\text{model}}}$, such that

$$x + \sigma(xW_{\text{read}})W_{\text{write}} = \sigma(xW'_{\text{read}})W'_{\text{write}} \qquad \text{for all } x \in \mathbb{R}^{1 \times d_{\text{model}}}. \tag{5}$$

Lemma A.1 (Invertible reduction)**.** *Let $M \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ be invertible. Then*

$$xM + \sigma(xW_{\text{read}})W_{\text{write}} = \sigma(xW'_{\text{read}})W'_{\text{write}} \quad \textit{for all } x$$

if and only if

$$x + \sigma(xW_{\text{read}})W_{\text{write}}M^{-1} = \sigma(xW'_{\text{read}})W'_{\text{write}}M^{-1} \quad \textit{for all } x.$$

Proof. Right-multiply by M^{-1}, which is invertible. □

Any invertible linear skip therefore reduces to the identity skip.

Proposition A.2 (Homogeneous blocks)**.** *Let B and B' be blocks that are positively homogeneous of degree $k \neq 1$, of any widths, and let $M \neq 0$. Then $xM + B(x) = B'(x)$ fails for some x.*

Examples are ReLU^2 and ReGLU blocks without biases, both of degree 2.

Proof. Outline. We compare the two sides along rays through the origin: a term of degree one and a term of degree $k \neq 1$ cannot agree at every scale.

Suppose it holds for all x and evaluate at λx with $\lambda > 0$: $\lambda xM + \lambda^k B(x) = \lambda^k B'(x)$. Dividing by λ gives $xM = \lambda^{k-1}\big(B'(x) - B(x)\big)$, and $\lambda^{k-1} \to 0$ as $\lambda \to 0$ if $k > 1$ and as $\lambda \to \infty$ if $k < 1$. So $xM = 0$ for all x, which contradicts $M \neq 0$. □

Proposition A.3 (Gated blocks)**.** *Let B and B' be gated blocks without biases, of any widths, whose gates are differentiable at 0 and vanish there, and let $M \neq 0$. Then $xM + B(x) = B'(x)$ fails for some x.*

Examples are SwiGLU and GeGLU blocks, which Shazeer (2020) defines without biases.

Proof. Outline. Near the origin a gated block is of second order in x, while xM is of first order.

A unit of a gated block computes $g(u \cdot x)\,(v \cdot x)$. Since $g(0) = 0$ and g is differentiable at 0, $|g(u \cdot x)| \le c\,\|x\|$ for small $\|x\|$, so the unit is $O(\|x\|^2)$, and so are B and B'. If $xM + B(x) = B'(x)$ for all x, then $xM = O(\|x\|^2)$ as $x \to 0$. Along a ray $x = tz$ this reads $t\,zM = O(t^2)$, so $zM = 0$ for every z, which contradicts $M \neq 0$. □

Corollary A.4 (Any depth)**.** *Let every block be as in Proposition A.2 with $k > 1$ or as in Proposition A.3, of any widths. Let F be a composition of residual blocks $h \mapsto h + B_\ell(h)$ and H a composition of one or more blocks without skips. Then $F \neq H$. With embeddings and readouts, the residual network $x \mapsto F(xE)D$ differs from the skipless network $x \mapsto H(xE')D'$ whenever $ED \neq 0$, also with pre-RMSNorm and $\varepsilon > 0$. Without normalization the same holds for* tanh *blocks.*

Proof. Each such block maps 0 to 0 and has Jacobian 0 there, because it is $O(\|x\|^k)$ with $k > 1$, respectively $O(\|x\|^2)$. By the chain rule F has Jacobian I at the origin and H has Jacobian 0; with embeddings and readouts these become ED and 0. Under pre-RMSNorm with $\varepsilon > 0$, $N(0) = 0$ and N has Jacobian $I/\sqrt{\varepsilon}$ at 0, so every block, read through N, still has Jacobian 0 there, and the residual and the skipless network have Jacobians $ED/\sqrt{\varepsilon}$ and 0 at the origin. For tanh blocks without normalization, H and $F(h) - h$ are bounded, so $F(xE)D - H(xE')D'$ differs from xED by a bounded function and does not vanish when $ED \neq 0$. □

Parity. The remaining activations have an odd part that is exactly linear. For ReLU, $\mathrm{ReLU}(z) = |z|/2 + z/2$. For GELU, $\mathrm{GELU}(z) = z\Phi(z)$ with Φ the standard normal distribution function, and since $\Phi(-z) = 1 - \Phi(z)$,

$$\mathrm{GELU}(z) = \Psi(z) + z/2, \qquad \Psi(z) := z\big(\Phi(z) - \tfrac{1}{2}\big), \qquad \Psi(-z) = \Psi(z).$$

The even parts, $|z|/2$ and Ψ, match the hidden units of the two blocks; the odd part, $z/2$, turns a sign flip into a linear term and gives the absorption condition.

Theorem A.5 (Identity absorption for ReLU)**.** *Let $\sigma = \mathrm{ReLU}$, $d_{\mathrm{model}} \ge 2$, and let the block be nondegenerate. Then (5) has a solution if and only if*

$$W_{\mathrm{read}}[:, S]\, W_{\mathrm{write}}[S, :] = -I_{d_{\mathrm{model}}} \quad \textit{for some } S. \tag{6}$$

Proof. Outline. Sufficiency is a construction: flipping the signs of the hidden units indexed by S produces the missing identity term. For necessity we split (5) into its odd and even parts in x, use the kinks of the even part to match the hidden units of the two blocks one to one, and read the index set S off the signs of the matching.

Sufficiency. Let $F = I_m - 2\Pi_S$, a diagonal sign matrix, and set $W'_{\mathrm{read}} = W_{\mathrm{read}}F$, $W'_{\mathrm{write}} = W_{\mathrm{write}}$. Since $|xW_{\mathrm{read}}F| = |xW_{\mathrm{read}}|$ coordinatewise,

$$\mathrm{ReLU}(xW'_{\mathrm{read}})W'_{\mathrm{write}} = \tfrac{1}{2}\big(xW_{\mathrm{read}}F + |xW_{\mathrm{read}}|\big)W_{\mathrm{write}},$$
$$x + \mathrm{ReLU}(xW_{\mathrm{read}})W_{\mathrm{write}} = \tfrac{1}{2}\big(xW_{\mathrm{read}} + |xW_{\mathrm{read}}|\big)W_{\mathrm{write}} + x,$$

and the two agree if and only if $\frac{1}{2}xW_{\mathrm{read}}(F - I_m)W_{\mathrm{write}} = x$ for all x, that is $-W_{\mathrm{read}}\Pi_S W_{\mathrm{write}} = I_{d_{\mathrm{model}}}$, which is (6).

Necessity. Suppose (5) holds and split both sides into odd and even parts in x:

$$W'_{\mathrm{read}}W'_{\mathrm{write}} = W_{\mathrm{read}}W_{\mathrm{write}} + 2I_{d_{\mathrm{model}}}, \qquad |xW'_{\mathrm{read}}|\, W'_{\mathrm{write}} = |xW_{\mathrm{read}}|\, W_{\mathrm{write}} \ \text{ for all } x. \tag{7}$$

Step 1, alignment of kinks. The map $x \mapsto |xW_{\mathrm{read}}|\, W_{\mathrm{write}}$ fails to be differentiable exactly on the union of the hyperplanes $\{xw_i = 0\}$; these are m distinct hyperplanes, each carrying a genuine kink since $\hat{w}_i \neq 0$. By the second identity in (7) the map $x \mapsto |xW'_{\mathrm{read}}|\, W'_{\mathrm{write}}$ has the same non-differentiability locus. Each of the m kink hyperplanes $\{x : xw_i = 0\}$ of the target needs at least one unit of B' whose kink lies on it, and B' has only m units; so each hyperplane gets exactly one unit, no two units share a hyperplane, and no kinks cancel. After relabeling, $W'_{\mathrm{read}} = W_{\mathrm{read}}C$ with

$C = \operatorname{diag}(c_1, \ldots, c_m)$, all $c_i \neq 0$. *Step 2, coefficients.* Substituting, $|xW_{\text{read}}C| = |xW_{\text{read}}|\,|C|$, so $|xW_{\text{read}}|\,|C|\,W'_{\text{write}} = |xW_{\text{read}}|\,W_{\text{write}}$ for all x. For $d_{\text{model}} \geq 2$ and pairwise non-collinear columns, the functions $|xw_i|$ are linearly independent modulo linear functions, because their kink hyperplanes are distinct; hence $|C|\,W'_{\text{write}} = W_{\text{write}}$ and $W'_{\text{write}} = |C|^{-1}W_{\text{write}}$. *Step 3, signs.* The first identity in (7) becomes $W_{\text{read}}C|C|^{-1}W_{\text{write}} = W_{\text{read}}W_{\text{write}} + 2I_{d_{\text{model}}}$. With $\Sigma = C|C|^{-1} = \operatorname{diag}(\operatorname{sign} c_i)$ this reads $W_{\text{read}}(\Sigma - I_m)W_{\text{write}} = 2I_{d_{\text{model}}}$; the diagonal of $\Sigma - I_m$ has entries 0 or -2, so with $S = \{i : c_i < 0\}$ we get $W_{\text{read}}\Pi_S W_{\text{write}} = -I_{d_{\text{model}}}$, and $|S| \geq d_{\text{model}}$ by the rank of $I_{d_{\text{model}}}$. □

The GELU case needs one classical fact about ridge functions, whose short proof we include.

Lemma A.6 (Vanishing sums of ridge functions)**.** *Let $a_1, \ldots, a_r \in \mathbb{R}^{d_{\text{model}}}$ be nonzero and pairwise non-collinear, and let $g_1, \ldots, g_r : \mathbb{R} \to \mathbb{R}^k$ be $r-1$ times continuously differentiable with $\sum_i g_i(a_i \cdot x) = 0$ for all $x \in \mathbb{R}^{d_{\text{model}}}$. Then every g_i is a polynomial of degree at most $r-2$.*

Proof. Fix i. For each $j \neq i$ choose $u_j \in \mathbb{R}^{d_{\text{model}}}$ with $u_j \cdot a_j = 0$ and $u_j \cdot a_i \neq 0$, which exists because a_i is not a multiple of a_j. The directional derivative ∂_{u_j} annihilates every function of $a_j \cdot x$ and maps a function of $a_k \cdot x$ to a function of $a_k \cdot x$. Applying the product of the commuting operators ∂_{u_j}, $j \neq i$, to the sum therefore leaves $\prod_{j \neq i}(u_j \cdot a_i)\, g_i^{(r-1)}(a_i \cdot x) = 0$. Since $a_i \neq 0$, $a_i \cdot x$ takes every real value, so $g_i^{(r-1)} = 0$. □

Theorem A.7 (Identity absorption for GELU)**.** *Let $\sigma = \text{GELU}$ and let the block be nondegenerate. Then (5) has a solution if and only if (6) holds.*

Proof. Outline. The proof follows that of Theorem A.5. Linear independence of ridge functions replaces the kink argument, and one extra step shows that matched hidden units have the same scale.

Sufficiency. With $F = I_m - 2\Pi_S$, $W'_{\text{read}} = W_{\text{read}}F$ and $W'_{\text{write}} = W_{\text{write}}$, evenness of Ψ gives $\Psi(xW_{\text{read}}F) = \Psi(xW_{\text{read}})$, so

$$\begin{aligned}\text{GELU}(xW'_{\text{read}})W'_{\text{write}} &= \Psi(xW_{\text{read}})W_{\text{write}} + \tfrac{1}{2}xW_{\text{read}}FW_{\text{write}} \\ &= \text{GELU}(xW_{\text{read}})W_{\text{write}} - xW_{\text{read}}\Pi_S W_{\text{write}} \\ &= \text{GELU}(xW_{\text{read}})W_{\text{write}} + x.\end{aligned}$$

Necessity. Splitting (5) into even and odd parts gives

$$\Psi(xW'_{\text{read}})W'_{\text{write}} = \Psi(xW_{\text{read}})W_{\text{write}} \ \text{ for all } x, \qquad W'_{\text{read}}W'_{\text{write}} = W_{\text{read}}W_{\text{write}} + 2I_{d_{\text{model}}}. \tag{8}$$

Step 1, alignment of directions. Ψ is analytic and not a polynomial. Zero columns of W'_{read} contribute $\Psi(0) = 0$; group the other columns of W'_{read} and the columns of W_{read} into classes of collinear vectors. In the first identity of (8) the terms of each class form one ridge function in the direction of the class, and these ridge functions sum to zero, so each is a polynomial by Lemma A.6. The class of a column w_i of W_{read} contains no other column of W_{read}, so if it contained no column of W'_{read} its ridge function would be $-\Psi(xw_i)\,\hat{w}_i$, which is not a polynomial because $\hat{w}_i \neq 0$. Hence each of the m classes of the columns of W_{read} contains a nonzero column of W'_{read}, and since W'_{read} has only m columns, after relabeling $W'_{\text{read}} = W_{\text{read}}C$ with $C = \operatorname{diag}(c_i)$, $c_i \neq 0$. *Step 2, unit scale.* Collecting the terms of direction w_i on both sides gives, for each i, a ridge function $g_i(xw_i) := \Psi(|c_i|\,xw_i)\,(W'_{\text{write}})_{i,:} - \Psi(xw_i)\,(W_{\text{write}})_{i,:}$ with $\sum_i g_i(xw_i) = 0$. By Lemma A.6 each g_i is a polynomial. For $t \to +\infty$, $\Psi(t) = t/2 - t\,\bar{\Phi}(t)$ with $\bar{\Phi}(t) = 1 - \Phi(t)$ exponentially small, so $g_i(t) = \frac{t}{2}\big(|c_i|(W'_{\text{write}})_{i,:} - (W_{\text{write}})_{i,:}\big) + o(1)$; a polynomial with this behavior is linear, and it is even because Ψ is even, so it is constant, and letting $t \to 0$ shows the constant is zero. Hence $|c_i|(W'_{\text{write}})_{i,:} = (W_{\text{write}})_{i,:}$ and the exponentially small remainder $|c_i|\,t\,\bar{\Phi}(|c_i|t)\,(W'_{\text{write}})_{i,:} - t\,\bar{\Phi}(t)\,(W_{\text{write}})_{i,:}$ must vanish, which with $(W_{\text{write}})_{i,:} \neq 0$ forces $\bar{\Phi}(|c_i|t) = \bar{\Phi}(t)$ for all $t > 0$, so $|c_i| = 1$. Therefore $|C| = I_m$ and $W'_{\text{write}} = W_{\text{write}}$. *Step 3, signs.* Identical to the ReLU case: the second identity in (8) becomes $W_{\text{read}}(C - I_m)W_{\text{write}} = 2I_{d_{\text{model}}}$ with $C = \operatorname{diag}(\pm 1)$, so $S = \{i : c_i = -1\}$ satisfies (6) and $|S| \geq d_{\text{model}}$. □

Remark A.8. Any activation whose even part ψ is analytic and not a polynomial matches the hidden units in the same way: Steps 1 and 2 of the proof of Theorem A.7 apply with ψ in place of Ψ (zero columns of W'_{read} now contribute a constant, which is a polynomial), except that the unit scale in Step 2 follows from a general fact: if $\sum_k a_k \psi(\alpha_k t)$ is a polynomial for distinct $\alpha_k > 0$, then every $a_k = 0$. Indeed, the Taylor coefficients $\psi_n \sum_k a_k \alpha_k^n$ vanish for all large n, infinitely many ψ_n are nonzero, and a generalized Vandermonde matrix with distinct positive nodes is invertible. Linear independence of the ridge functions alone does not suffice: for $\sigma(z) = z/2 + |z|^3$ the even part is homogeneous, so a unit can be rescaled together with its output row, and the unit scale is not forced. The odd part o of σ enters only through the flips: with the units matched, both directions reduce to $x = -2 \sum_{i \in S} o(x w_i)\, \hat{w}_i$, with S the flipped units. For $o(z) = z/2$, that is $\sigma(z) - \sigma(-z) = z$, as for SiLU, whose even part is $\frac{z}{2} \tanh \frac{z}{2}$, this is (6), and $o(z) = \alpha z$ with $\alpha \neq 0$ only rescales it; a polynomial o of higher degree adds the condition that the units in S cancel its terms of degree three and higher, and a smooth o that is not a polynomial admits no S (Lemma A.6), so the skip is never absorbed at the same width.

Remark A.9 (Singular linear skips). Nothing in the proofs of Theorems A.5 and A.7 uses that the linear term is the identity. With xM in place of x on the left of (5), the odd part of the equation reads $W'_{\text{read}} W'_{\text{write}} = W_{\text{read}} W_{\text{write}} + 2M$, the even part is unchanged, and the sign step yields $W_{\text{read}} \Pi_S W_{\text{write}} = -M$. So for any $M \neq 0$, invertible or not, absorption holds if and only if $W_{\text{read}}[:, S] W_{\text{write}}[S, :] = -M$ for some S, with $|S| \geq \operatorname{rank} M$, which is a measure-zero condition on the weights for each fixed M. This gives Theorem 3.1(i) for singular M.

Corollary A.10 (Generic disjointness for one block)**.** *Let σ be ReLU or GELU and $d_{\text{model}} \geq 2$. For generic weights,* (5) *has no solution.*

Proof. Outline. The condition of Theorems A.5 and A.7 is a finite union of algebraic sets of positive codimension, and such a union has measure zero.

For each of the finitely many S, (6) is d_{model}^2 polynomial equations in the weights and defines a set of measure zero; the hypotheses of Theorems A.5 and A.7 hold outside another set of measure zero. So a generic residual block is not a block of the same width without a skip, and, by the same argument with the two sides of (5) exchanged, a generic block without a skip is not a residual block of the same width: neither class contains the other. □

Proposition A.11 (d_{model} more units)**.** *Let σ be ReLU or GELU, $d_{\text{model}} \geq 2$ and $M \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ invertible.*
(i) *If W_{read} has rank d_{model}, a block B' of width $m + d_{\text{model}}$ satisfies $xM + \sigma(xW_{\text{read}})W_{\text{write}} = B'(x)$ for all x.*
(ii) *For generic weights no block of width less than $m + d_{\text{model}}$ does.*

Proof. Outline. Two units with opposite reads produce a linear term of rank one, so d_{model} extra units produce xM. Conversely, the matching of the units in Theorems A.5 and A.7 shows that every extra unit adds at most one such term.

(i) As $\sigma(z) - \sigma(-z) = z$, a column w and rows a, b give $\sigma(xw)(a + b) + \sigma(-xw)(-b) = \sigma(xw)\, a + (xw)\, b$. Choose d_{model} units whose columns $w_{i_1}, \ldots, w_{i_{d_{\text{model}}}}$ form an invertible matrix W_K, and let $b_1, \ldots, b_{d_{\text{model}}}$ be the rows of $W_K^{-1} M$. Adding b_k to the row $\hat{w}_{i_k}$, and a unit with column $-w_{i_k}$ and row $-b_k$, for each k, gives a block of width $m + d_{\text{model}}$ that computes $\sigma(xW_{\text{read}})W_{\text{write}} + xW_K W_K^{-1} M$.

(ii) Outside a set of measure zero the block is nondegenerate, and for each of the finitely many sets S of units the determinant of $M + W_{\text{read}}[:, S] W_{\text{write}}[S, :]$ is nonzero, since it is a polynomial in the weights equal to $\det M \neq 0$ at $W_{\text{write}} = 0$. Let a block B' of width m', with columns w'_j and rows $\hat{w}'_j$, satisfy $xM + B(x) = B'(x)$. Its odd part gives $W'_{\text{read}} W'_{\text{write}} = W_{\text{read}} W_{\text{write}} + 2M$, and its even part $|xW'_{\text{read}}|\, W'_{\text{write}} = |xW_{\text{read}}|\, W_{\text{write}}$ for ReLU and $\Psi(xW'_{\text{read}})W'_{\text{write}} = \Psi(xW_{\text{read}})W_{\text{write}}$ for GELU. Group the nonzero columns w'_j and the columns w_i into classes of collinear vectors, write $w'_j = s_j u$ with a unit vector u of the class, $u = w_i / \|w_i\|$ in the class of w_i, and let $P_u = \|w_i\| \hat{w}_i$ there and $P_u = 0$ in a class without a column of W_{read}. The even part then forces, class by class, $\sum_j |s_j|\, \hat{w}'_j = P_u$ for ReLU, since the functions $|xu|$ of distinct classes are linearly independent

(Step 2 of Theorem A.5), and for GELU, by Lemma A.6 and the independence of the functions $\Psi(\alpha t)$ for distinct $\alpha > 0$ modulo polynomials (Remark A.8), $\sum_{|s_j|=\alpha} \hat{w}'_j = \hat{w}_i$ if w_i lies in the class and $\alpha = \|w_i\|$, and 0 otherwise. With $Q_u = \sum_j s_j \hat{w}'_j$ the odd part reads $\sum_u u\,(Q_u - P_u) = 2M$. Call a class mixed if it contains w'_j of both signs, for GELU of both signs and equal $|s_j|$. A class that is not mixed has $Q_u = P_u$ or $Q_u = -P_u$; let S collect the units i whose class is not mixed and has $Q_u = -P_u \neq 0$. Then

$$2\big(M + W_{\text{read}}[:, S] W_{\text{write}}[S, :]\big) = \sum_{u \text{ mixed}} u\,(Q_u - P_u).$$

Every class of a column w_i contains some w'_j, since $P_u \neq 0$, and every mixed class contains at least two, so $m' \geq m$ plus the number of mixed classes. If $m' < m + d_{\text{model}}$, the right side has rank less than d_{model}, so the left side is singular, which the weights exclude. □

Proof of Theorems 3.2 and 4.1. For homogeneous and gated blocks, Theorem 3.2 is Propositions A.2 and A.3, and Theorem 4.1(i) is Corollary A.4. For $\tanh$ blocks, B_θ and $B_{\theta'}$ are bounded and xM is not, which gives Theorem 3.2, and Theorem 4.1(ii) is again Corollary A.4. □

Proof of Theorem 3.1. Part (i) is Theorems A.5 and A.7, with Lemma A.1 for invertible M and Remark A.9 for singular M; for each M the condition holds only on a set of weights of measure zero (Corollary A.10 and Remark A.9). Part (ii) is Proposition A.11. □

Proof of Theorem 4.2. MLP blocks act on each token separately, so we compare the networks on rows $x \in \mathbb{R}^{1 \times d_{\text{in}}}$. Outside a set of weights of measure zero, E and D have rank d_{model}, no read of a unit vanishes, and $I + W^\ell_{\text{read}}[:, S]\, W^\ell_{\text{write}}[S, :]$ is invertible for every block ℓ and every set S of its units, since each determinant is a polynomial in the weights that equals 1 when the writes vanish. On an open set of inputs on which no unit of the residual network switches, with active sets S_ℓ, the network is the linear map $x \mapsto xEMD$ with $M = \prod_\ell \big(I + W^\ell_{\text{read}}[:, S_\ell]\, W^\ell_{\text{write}}[S_\ell, :]\big)$ invertible, of rank d_{model}. By induction over the blocks, on such a set the pre-activation of each unit of the next block is x times a nonzero vector, so the inputs at which some unit switches have measure zero, and the inputs near which the network is linear of rank d_{model} form an open dense set Ω.

Let a skipless network F' have embedding E' and a first block with reads $w'_1, \ldots, w'_{m'}$, $m' < 2d_{\text{model}}$, and set $a_i = E' w'_i$. Take x_0 with $x_0, -x_0 \in \Omega$ and $x_0 a_i \neq 0$ whenever $a_i \neq 0$. The units active near x_0, those with $x_0 a_i > 0$, and those active near $-x_0$ are disjoint, so one of the two sets, say the first after replacing x_0 by $-x_0$ if needed, has fewer than d_{model} units. Near x_0 the other units of the first block output zero, so F' depends on x only through the numbers xa_i of the active units, and $F'(x_0 + u) = F'(x_0)$ for every small u in a subspace of dimension greater than $d_{\text{in}} - d_{\text{model}}$. The residual network F changes by $uEMD$, which vanishes only on a subspace of dimension $d_{\text{in}} - d_{\text{model}}$. So some small u changes F and leaves F' unchanged, and $F' \neq F$. □

Proof of Proposition 4.3. Theorem 3.1(ii) with $M = I$ turns each residual block, whose read has rank d_{model}, into a block without a skip with d_{model} more hidden units. Composing these blocks, with the same embedding and readout, gives a skipless network that equals the residual one. □

A.2 No absorption in attention (Theorem 3.3, Proposition A.14)

Theorem 3.3 collects what the main text uses of the following theorem, for the attention blocks B_θ of (4).

Theorem A.12 (Attention and linear terms)**.** *Let θ and θ' be parameters of attention blocks with any numbers and widths of heads, and let $M \neq 0$ be a $d_{\text{model}} \times d_{\text{model}}$ matrix.*
(i) *For almost every $h \in \mathbb{R}^{T \times d_{\text{model}}}$, $hM + B_\theta(h) \neq B_{\theta'}(h)$.*
(ii) *Some h whose rows have norm at most 1 has a row in which $hM + B_\theta(h)$ and $B_{\theta'}(h)$ differ by at least $\|M\|/3$, where $\|M\|$ is the spectral norm.*
(iii) *With an embedding E and a readout D with $ED \neq 0$, the residual network $x \mapsto \big(xE + B_\theta(xE)\big)D$ differs from every skipless network $x \mapsto B_{\theta'}(xE')D'$.*

Overview. Two kinds of input suffice. If all tokens are equal, attention returns them unchanged, so the value–output products of all heads must add up to M. If one token is zero, all its scores are zero, so every head gives it the average of the tokens, while the skip gives it zero; this asks the same sum to vanish. Part (ii) makes the conflict quantitative, and analyticity turns it into part (i). With an embedding and a readout, the same two inputs give part (iii).

Proof of Theorem A.12. Let $F(h) = hM + B_\theta(h) - B_{\theta'}(h)$, and let θ' have H' heads. The difference $B_{\theta'} - B_\theta$ is one attention block with $H + H'$ heads,

$$B_{\theta'}(h) - B_\theta(h) = \sum_{k=1}^{H+H'} R_k(h)\, h\, C_k,$$

where the $R_k(h)$ are the attention maps of the heads of θ' and of θ, and the C_k are their value–output products, with a minus sign for the heads of θ. Let $C = \sum_k C_k$.

Step 1: equal tokens. Let every row of h be x. Each row of $R_k(h)$ sums to 1, so every row of $R_k(h)h$ is x. Hence every row of $F(h)$ is $x(M - C)$.

Step 2: one zero token. Let the last row of h be 0 and the other rows y. Every score in the last row of $R_k(h)$ is a product with the last row of h, which is 0. So that row of $R_k(h)$ is $(1/T, \dots, 1/T)$, also with a causal mask, since the last token sees all T tokens. Hence the last row of $F(h)$ is $-\frac{T-1}{T}\, yC$.

Proof of (ii). Over $\|x\|, \|y\| \le 1$, Step 1 gives rows of norm up to $\|M - C\|$, and Step 2 up to $\frac{T-1}{T}\|C\| \ge \frac{1}{2}\|C\|$. If $\|M - C\| < \|M\|/3$, then $\|C\| > \frac{2}{3}\|M\|$, so Step 2 reaches more than $\|M\|/3$.

Proof of (i). F is real-analytic, because the softmax is, and by (ii) some entry of F is not identically zero. The zeros of a real-analytic function that is not identically zero form a set of measure zero: in one variable they are isolated, and the general case follows by induction on the dimension with Fubini's theorem.

Proof of (iii). Let C_θ and $C_{\theta'}$ be the sums of the value–output products of the heads of θ and of θ'. On inputs whose rows all equal a row vector u, Step 1 gives the rows $uE(I + C_\theta)D$ in the residual network and $uE'C_{\theta'}D'$ in the network without a skip. On inputs whose last row is 0 and whose other rows equal v, Step 2 gives the last rows $\frac{T-1}{T}\, vEC_\theta D$ and $\frac{T-1}{T}\, vE'C_{\theta'}D'$. If the two networks are equal, then $E(I + C_\theta)D = E'C_{\theta'}D' = EC_\theta D$, so $ED = 0$. □

We recall Theorem 3.3 from Section 3.

Theorem 3.3 (Attention blocks)**.** *Let θ and θ' be parameters of attention blocks with any numbers and widths of heads, and let $M \neq 0$ be a $d_{\text{model}} \times d_{\text{model}}$ matrix.*
(i) *For almost every $h \in \mathbb{R}^{T \times d_{\text{model}}}$, $hM + B_\theta(h) \neq B_{\theta'}(h)$.*
(ii) *With an embedding E and a readout D with $ED \neq 0$, the residual network $x \mapsto \big(xE + B_\theta(xE)\big)D$ differs from every skipless network $x \mapsto B_{\theta'}(xE')D'$.*

Proof of Theorem 3.3. Parts (i) and (ii) are Theorem A.12(i) and (iii). □

Proposition A.14 follows from a lemma on attention heads.

Lemma A.13 (Attention heads are identifiable)**.** *Let $T \ge 3$. Let $\Gamma_1, \dots, \Gamma_m \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ be distinct, let $C_1, \dots, C_m, M \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$, and write $P_\Gamma(h) = \operatorname{softmax}(h\Gamma h^\top)$, with the softmax taken row by row. If*

$$hM + \sum_{k=1}^{m} P_{\Gamma_k}(h)\, h\, C_k = 0$$

for all h in a nonempty open subset of $\mathbb{R}^{T \times d_{\text{model}}}$, then $M = 0$ and $C_1 = \cdots = C_m = 0$.

Overview. Move one token along a complex line. Each head becomes a meromorphic function of one complex variable, whose poles lie on a vertical line that its score matrix determines, so heads with distinct score matrices have disjoint poles. At a pole of one head the other terms are regular, so its residue must vanish, and the residues force its value–output product to vanish. What is left is the skip and a uniform head, which two simple inputs separate.

Proof of Lemma A.13. One moving token. Fix h in the open set, with rows $x_1, \dots, x_T$, and fix $v \in \mathbb{R}^{1\times d_{\text{model}}}$. Replace the second row by $x_2 + tv$ with $t \in \mathbb{C}$, and look at the first row. For a head with score matrix Γ, the first row has scores $z_r = x_1 \Gamma x_r^\top$ for $r \neq 2$, and $a + bt$ for $r = 2$, where $a = x_1 \Gamma x_2^\top$ and $b = x_1 \Gamma v^\top$. Its output is

$$g_\Gamma(t) = \frac{e^{a+bt}\,(x_2 + tv) + Z\bar{x}}{e^{a+bt} + Z}, \qquad Z = \sum_{r\neq 2} e^{z_r}, \qquad \bar{x} = \frac{1}{Z}\sum_{r\neq 2} e^{z_r} x_r,$$

where $\bar{x}$ is the average of the other tokens under the weights of the head. The first row of hM is $x_1 M$, which does not depend on t.

Poles. If $b \neq 0$, then g_Γ is meromorphic on $\mathbb{C}$, with simple poles at

$$t_n = c + \frac{(2n+1)\pi i}{b}, \quad n \in \mathbb{Z}, \qquad \text{where } c = \frac{\log Z - a}{b},$$

so all its poles lie on the vertical line $\operatorname{Re} t = c$. The residue at t_n is $(x_2 + t_n v - \bar{x})/b$. If $\Gamma = 0$, then g_Γ is affine in t and has no poles.

The poles tell the heads apart. For h and v outside a closed set with empty interior, $b \neq 0$ for every $\Gamma_k \neq 0$, and distinct $\Gamma_k \neq 0$ give distinct lines $\operatorname{Re} t = c$. The first condition fails only on the zero sets of nonzero polynomials. For the second, suppose that two heads $\Gamma \neq \Gamma'$ have $c = c'$ for all h and v near some point. Clearing denominators and comparing the terms bilinear in (x_2, v) gives $x_1\Gamma' = \lambda\, x_1\Gamma$ for some number $\lambda \neq 0$. Comparing the terms linear in v then gives $\log Z' = \lambda \log Z$, that is,

$$\log \sum_{r\neq 2} e^{\lambda z_r} \;=\; \lambda \log \sum_{r\neq 2} e^{z_r}.$$

Because $T \geq 3$, the sum has at least two terms, $r = 1$ and $r = 3$, and z_3 varies with x_3. Differentiating in z_3 gives $\operatorname{softmax}(\lambda z)_3 = \operatorname{softmax}(z)_3$ on an interval of z_3, which forces $\lambda = 1$. So $x_1\Gamma' = x_1\Gamma$ for all x_1 near a point, and $\Gamma' = \Gamma$, a contradiction. With two tokens the sum has the single term $r = 1$, and this step fails (Section 3).

Residues. The identity holds for real t near 0, so by analytic continuation it holds on all of $\mathbb{C}$ away from the poles. At a pole t_n of a head $\Gamma_k \neq 0$, no other term is singular, so the residue times C_k vanishes: $(x_2 + t_n v - \bar{x})\, C_k = 0$ for every $n \in \mathbb{Z}$. Subtracting two consecutive ones gives $vC_k = 0$. This holds for all v in an open set, so $C_k = 0$.

The skip. What remains is $hM + P_0(h)\, h\, C_0 = 0$, where C_0 is the coefficient of $\Gamma = 0$ (or 0 if no Γ_k vanishes). This is linear in h, so it holds for all h. Take the first row of h equal to x and the other rows 0: the first two rows give $x(M + C_0/T) = 0$ and $xC_0/T = 0$. So $C_0 = 0$ and $M = 0$. □

The lemma gives the proposition that Section 3.3 cites: with three or more tokens, the heads must cancel in groups with the same score matrix.

Proposition A.14 (The softmaxes must match)**.** *Let $T \geq 3$, let θ and θ' be parameters of attention blocks, and suppose that $hM + B_\theta(h) = B_{\theta'}(h)$ for all h in a nonempty open set. Then $M = 0$, and for every matrix Γ, the heads with score matrix Γ have the same sum of value–output products in B_θ as in $B_{\theta'}$.*

Proof of Proposition A.14. Group the heads of $B_{\theta'}$ and B_θ by score matrix, add their value–output products, with a minus sign for the heads of θ, and apply Lemma A.13. □

With two tokens, heads can cancel: the softmax of two scores is the logistic function of their difference, and $1/(1+e^{-z}) + 1/(1+e^{z}) = 1$ makes two heads with opposite score matrices add up to a uniform head. A skip still cannot be absorbed when $T = 2$ (Theorem 3.3(i)).

A.3 Absorption under RMSNorm with $\varepsilon = 0$ (Proposition 4.4)

We recall Proposition 4.4 from Section 4.

Proposition 4.4 (Absorption under RMSNorm with $\varepsilon = 0$)**.** *Consider pre-RMSNorm networks with $\varepsilon = 0$, with blocks of any kind and width.*
(i) *The function F of every skipless network satisfies $F(cx) = F(x)$ for all $c > 0$.*
(ii) *A residual network with this property computes $x \mapsto N(xE)D$.*
(iii) *The residual networks that equal a skipless network have parameters in a set of measure zero.*

Proof. (i) The skipless network reads its input only through $N(xE)$, and $N(cxE) = N(xE)$ for $c > 0$ when $\varepsilon = 0$.

(ii) Fix an input x whose rows of xE are nonzero, and let $c \to \infty$. Every block reads rows of length $\sqrt{d_{\text{model}}}$, so its output stays bounded, and the hidden states of the residual network at cx are $h_\ell = cxE + O(1)$. Hence $N(h_L) \to N(xE)$ row by row, and $F(cx) \to N(xE)D$. If $F(cx) = F(x)$ for all $c > 0$, then $F(x) = N(xE)D$.

(iii) By (i) and (ii), a residual network that equals a skipless network computes $x \mapsto N(xE)D$; we show that the set Z of such parameters has measure zero. Outside a set of measure zero, E and D have rank d_{model}, so $vD = v'D$ only for $v = v'$. Block ℓ adds $a_\ell(x)W^\ell_{\text{write}}$ to the stream, where $a_\ell(x)$ is the input of its write: the activations of its hidden units at $N(h_{\ell-1})$, and for attention the rows $P_{\Gamma_i}(N(h_{\ell-1}))\, N(h_{\ell-1})\, W_{V,i}$ of its heads, with the output projections as the write. If a_ℓ is not identically zero but every a_k with $k > \ell$ is, the blocks after ℓ add nothing, so the parameters lie in the set Y_ℓ where a_ℓ is not identically zero and $N(h_\ell(x)) = N(xE)$ for all x. Fix all parameters but W^ℓ_{write}, which leaves $h_{\ell-1}$ and a_ℓ unchanged, and an input x with a row t in which $a_\ell(x) \neq 0$. In that row the condition asks $a_{\ell,t}(x)W^\ell_{\text{write}}$ to lie on the ray $\{\lambda x_t E - h_{\ell-1,t}(x) : \lambda > 0\}$. As W^ℓ_{write} varies, $a_{\ell,t}(x)W^\ell_{\text{write}}$ takes every value in $\mathbb{R}^{1\times d_{\text{model}}}$, so this happens on a set of measure zero, since $d_{\text{model}} \geq 2$, and by Fubini's theorem Y_ℓ has measure zero. On the rest of Z every a_ℓ vanishes identically, in particular a_1, whose argument $N(xE)$ takes every value on the sphere of radius $\sqrt{d_{\text{model}}}$. For each block of Section 2 this forces a column of the read of block 1, or of one of its value matrices, to vanish, which again happens on a set of measure zero. □

A.4 Skipless networks as limits of residual networks (Proposition 5.1)

Proposition 5.1 (Skipless networks as limits of residual networks)**.** *Let F be a skipless network whose blocks are locally Lipschitz.*
(i) Without normalization. *The network F is a limit of residual networks with blocks of the same kinds and widths, uniformly on bounded sets of inputs.*
(ii) Under pre-RMSNorm with $\varepsilon = 0$. *If no hidden state of F has a zero row at any input x whose rows of xE are nonzero, the same holds on bounded sets of such inputs.*

Let F have the embedding E, the blocks B_ℓ, with reads W^ℓ_{read} and writes W^ℓ_{write}, and the readout D, and let $s_0 = xE$ and $s_\ell = B_\ell(s_{\ell-1})$ be its hidden states, so that $F(x) = s_L D$. For $t \geq 1$, let F_t be the residual network with the embedding E, the reads $t^{1-\ell}W^\ell_{\text{read}}$, the writes $t^\ell W^\ell_{\text{write}}$ and the readout $t^{-L}D$; under pre-RMSNorm with $\varepsilon = 0$, let F_t keep the embedding, the reads and the readout of F and have the writes $t^\ell W^\ell_{\text{write}}$. Let $r_0 = 0$ and

$$r_\ell \;=\; B_\ell(s_{\ell-1} + r_{\ell-1}) - B_\ell(s_{\ell-1}) \;+\; t^{-1}\,(s_{\ell-1} + r_{\ell-1}). \tag{9}$$

The proofs show that the hidden states of F_t are $t^\ell(s_\ell + r_\ell)$ and bound r_ℓ, which shows that F_t tends to F. Norms are Frobenius norms for hidden states and operator norms for matrices.

Proof of (i). Block ℓ of F_t has its reads scaled by $t^{1-\ell}$ and its writes by t^ℓ, so it maps h to $t^\ell B_\ell(t^{1-\ell}h)$. We show by induction that its hidden states are $h_\ell = t^\ell(s_\ell + r_\ell)$, with $r_0 = 0$ and r_ℓ given by (9). For $\ell = 0$, $h_0 = xE = s_0$. If $h_{\ell-1} = t^{\ell-1}(s_{\ell-1} + r_{\ell-1})$, then

$$h_\ell \;=\; h_{\ell-1} + t^\ell B_\ell(t^{1-\ell}h_{\ell-1}) \;=\; t^\ell\big(B_\ell(s_{\ell-1} + r_{\ell-1}) + t^{-1}(s_{\ell-1} + r_{\ell-1})\big) \;=\; t^\ell(s_\ell + r_\ell).$$

Let X be bounded and $\sigma_\ell = \sup_{x\in X}\|s_\ell\|$, which is finite because the blocks are continuous. Suppose that $\|r_{\ell-1}\| \leq C_{\ell-1}/t$ for all $x \in X$ and $t \geq 1$, which holds for $\ell = 1$ with $C_0 = 0$. Then $s_{\ell-1}$ and

$s_{\ell-1} + r_{\ell-1}$ lie in the ball of radius $\sigma_{\ell-1} + C_{\ell-1}$, on which B_ℓ is Lipschitz with some constant Λ_ℓ, since B_ℓ is locally Lipschitz. Hence

$$\|r_\ell\| \;\le\; \Lambda_\ell \|r_{\ell-1}\| + t^{-1}\big(\sigma_{\ell-1} + \|r_{\ell-1}\|\big) \;\le\; C_\ell/t, \qquad C_\ell = \Lambda_\ell C_{\ell-1} + \sigma_{\ell-1} + C_{\ell-1}.$$

Finally $F_t(x) = t^{-L} h_L D = F(x) + r_L D$, so $\|F_t(x) - F(x)\| \le C_L \|D\|/t$ for all $x \in X$ and $t \ge 1$. □

Proof of (ii). Let X be a bounded set of inputs whose rows of xE are nonzero. Here N divides each row of h by its norm and multiplies it by $\sqrt{d_{\text{model}}}$, so $N(ch) = N(h)$ for $c > 0$, and $\|N(y') - N(y)\| \le 2\sqrt{d_{\text{model}}}\,\|y' - y\|/\|y\|$ for nonzero rows y, y'. The hidden states of F are $s_0 = xE$ and $s_\ell = B_\ell(N(s_{\ell-1}))$, and $F(x) = N(s_L)D$. For $\ell \ge 1$, s_ℓ depends on x only through $N(xE)$, whose rows lie on the sphere of radius $\sqrt{d_{\text{model}}}$; as no row of s_ℓ vanishes, continuity and compactness give $c > 0$ below the norm of every row of every s_ℓ, $\ell \ge 1$. Since N removes the factor $t^{\ell-1}$, the hidden states of F_t are again $h_0 = s_0$ and $h_\ell = t^\ell(s_\ell + r_\ell)$, now with $r_0 = 0$ and

$$r_\ell \;=\; B_\ell\big(N(s_{\ell-1} + r_{\ell-1})\big) - B_\ell\big(N(s_{\ell-1})\big) \;+\; t^{-1}\,(s_{\ell-1} + r_{\ell-1}),$$

as long as no row of $s_{\ell-1} + r_{\ell-1}$ vanishes. The maps B_ℓ are Lipschitz on the bounded image of N, with some constants Λ_ℓ, and $\sigma_\ell = \sup_{x\in X}\|s_\ell\|$ is finite. If every row of $r_{\ell-1}$ is shorter than c, then $\|r_\ell\| \le 2\sqrt{d_{\text{model}}}\,\Lambda_\ell\|r_{\ell-1}\|/c + t^{-1}(\sigma_{\ell-1} + \|r_{\ell-1}\|)$, so by induction $\|r_\ell\| \le C_\ell/t$ on X, and every row of r_ℓ is shorter than c once $t \ge t_0 = \max_\ell 2C_\ell/c$. Finally $\|F_t(x) - F(x)\| = \|(N(s_L + r_L) - N(s_L))D\| \le 2\sqrt{d_{\text{model}}}\,\|D\|\,C_L/(ct)$ for all $x \in X$ and $t \ge t_0$. □

Remark A.15 (The rate). The proofs give the rate $1/t$. Without normalization, for every bounded set X of inputs there is C such that $\|F_t(x) - F(x)\| \le C/t$ for all $x \in X$ and $t \ge 1$. Under pre-RMSNorm with $\varepsilon = 0$, the same holds for all $t \ge t_0$, for every bounded set X of inputs whose rows of xE are nonzero, with t_0 depending on X. The constants depend on X and on F.

B Universal approximation: width and biases

Lin and Jegelka (2018) show that residual ReLU networks whose blocks have a single hidden unit, $h \mapsto h + \mathrm{ReLU}(hw + c)\,v$ with $w \in \mathbb{R}^{d\times 1}$, $v \in \mathbb{R}^{1\times d}$ and a bias $c \in \mathbb{R}$, approximate every integrable function on $\mathbb{R}^d$ in L^1 as the depth grows, while ReLU networks without skips whose hidden layers have width at most the input dimension d cannot (their Proposition 2.1). The separation is confined to this narrow regime: with hidden layers of width $d + 1$, ReLU networks without skips already approximate every continuous function uniformly on compact sets (Hanin and Sellke, 2017).

The construction also needs the biases, for the shifts and the cut-offs at constants of its blocks (their Proposition 3.2). Without biases, a ReLU network, with or without skips, is positively homogeneous of degree one, $F(\lambda x) = \lambda F(x)$ for $\lambda > 0$, since ReLU and the linear maps are. On every ray $\{t\theta : t > 0\}$ such a network is a multiple of t, and so is every limit of such networks, uniformly or in L^1 on a compact set, on almost every ray: the multiples of t form a closed space, and by Fubini L^1 convergence passes to almost every ray. In one dimension, for instance, every such network is $x \mapsto a\,\mathrm{ReLU}(x) - b\,\mathrm{ReLU}(-x)$, which stays at L^1 distance at least $(2 - \sqrt{2})/3$ from x^2 on $[-1, 1]$. On all of $\mathbb{R}^d$, no integrable function other than 0 can be approximated in L^1, since a positively homogeneous function other than 0 is not integrable.